Dual Channel Bidirectional Brushed Electronic Speed Controller Manual

Characteristics

1. Dual-channel bidirectional differential output enables precise control of left and right tracks, allowing flexible steering and delivering an authentic off-road driving experience. Perfectly compatible with various RC tracked vehicles.

2. High-efficiency high-current UBEC power supply ensures stable voltage output for receivers and servos. The isolated design between MCU and UBEC significantly enhances system stability and noise immunity.

3. Multi-level safety protection ensures reliable and secure operation. Equipped with fail-safe and over-temperature protection functions, the system automatically responds to abnormal conditions, effectively preventing equipment damage and ensuring driving safety.

4. Low internal resistance MOSFET power transistors deliver efficient low-temperature operation with exceptional current load capacity. Maintains stable performance even under prolonged heavy-load conditions.

5. Linear throttle response provides refined and precise control. The optimized throttle curve enables smooth, linear power output with gentle starts and seamless acceleration, creating a highly authentic.

Specifications

Model	Continuous current	Burst current （10s）	LiPo Battery Cell	Dimension LxWxH （mm）	Weight （g）	Output		Braking mode
5Ax2	5A	13A	2S-3S	40x25X6.6	15.5	BEC	5V@500mA	NO
5Ax2Pro	5A	18A	2S-6S	50X25X5	20.5	BEC	5V@800mA	NO
10Ax2	10A	60A	2S-6S	50X25X5	22	BEC	5V@800mA	NO
20Ax2	20A	80A	2S-6S	62X41X10	58	UBEC	5V@3A	YES
30Ax2	30A	120A	2S-6S	62X41X10	58	UBEC	5V@3A	YES
40Ax2	40A	150A	2S-6S	62X48X10	66.5	UBEC	5V@3A	YES
50Ax2	50A	160A	3S-6S	72X52X10	70.5	UBEC	5V@3A	YES

Function Description

1. Throttle Calibration & Neutral Start

Default neutral position: 1500±50μs.

Auto throttle range detection: 900μs–2100μs.

Full compatibility with standard transmitters for quick setup.

2. LED Status Indicators

Solid Red: Power ON.

Green: Mixing Mode active (turns off when armed).

Blue: Independent Mode active (turns off when armed).

Clear visual feedback for easy operation.

3. Brake ON / Brake OFF Mode

Brake ON: Motor braking activates when throttle returns to neutral.

Brake OFF: Free motor coasting at neutral throttle.

Adjustable to match driving style and vehicle type.

Selectable modes allow for adaptation to various driving styles and vehicle applications.

4. Independent / Mixing Mod

Independent Mode: Separate control of dual motors.

Mixing Mode:

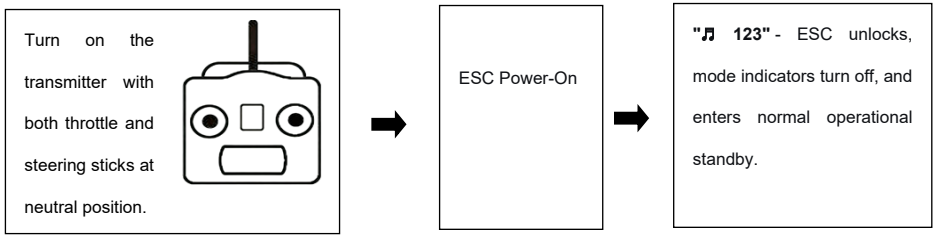
Throttle channel controls both motors simultaneously.

Steering channel adjusts differential power for precise turns.

Ideal for tracked vehicles and stable steering.

Flexible switching for various RC applications.

Normal startup Sequence



Important Notes:

Pre-Start Check: Before powering on the ESC, ensure both transmitter sticks are at neutral position and verify the receiver's two channels are securely connected to the dual signal cables. This guarantees proper initialization in either Independent or Mixing Mode.

Shutdown Sequence: Always disconnect the ESC from the power source before turning off the transmitter.

Protection Features

Fail-Safe (Signal Loss Protection)

Activation: Triggered upon signal loss

Action: Power ramps down to zero within 2 seconds, stopping the motor.

Recovery: After signal re-acquisition, the throttle must be returned to the neutral position to re-arm and resume normal operation.

Over-Temperature Protection

Threshold: Activates when the power MOSFET temperature exceeds approximately 95°C.

Action: Motor operation is halted.

Recovery: Normal operation automatically resumes once the temperature decreases to a safe level.

Troubleshooting

Trouble	Possible Causes	Solution
No motor response and power LED remains off after applying power.	1. Loose or faulty power connector. 2. Power switch is in the "OFF" position.	1. Reconnect or replace the power connector. 2. Toggle the power switch and observe the status indicator.
No motor response with mode LED solid on and ESC fails to arm after power-up.	1. Loose or faulty connection on the steering or throttle channel signal wire. 2. The steering or throttle stick/trigger on the transmitter is not returned to the neutral position.	1) Reseat the signal wire connectors to ensure proper contact. 2) Adjust the trim for the corresponding channel on your transmitter until its PWM signal is within the neutral zone.
After the mode switch, the working mode remains unchanged.	Mode change unsuccessful.	1) Check that the jumper is properly seated and making good contact. 2) Power cycle the ESC after changing the mode for the new setting to take effect.
Throttle and steering control channels are reversed.	1) The throttle and steering signal wires are connected to the wrong channels on the receiver. 2) Incorrect motor rotation direction.	1)Swap the throttle and steering signal cables at the receiver. 2)Reverse any two motor leads on the corresponding motor.